Design and Measured Behavior of a Hybrid Precast Concrete Wall Specimen for Seismic Regions

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Abstract: This paper presents the measured behavior from the testing of a 0.4-scale "hybrid" precast concrete wall specimen under reversed-cyclic lateral loading and provides an assessment of the seismic design and analysis of the wall by using the experimental results. The hybrid precast wall system investigated in the paper utilizes a combination of mild (i.e., Grade 400) steel and high-strength unbonded posttensioning (PT) steel for lateral resistance across horizontal joints. A seismic design procedure that conforms to ACI 318 and ACI ITG-5.2 was used for the design of the test specimen based on ACI ITG-5.1. The behavior of the specimen was measured with conventional data acquisition techniques and also full-field digital image correlation of the base panel and the critical joint between the base panel and the foundation, providing unprecedented information on the wall performance. The paper compares these measurements with the design and analytical predictions, focusing specifically on the applied lateral load and displacement, energy dissipation, behavior of the steel reinforcement, and behavior along the horizontal joints. The test specimen was not able to reach the expected ultimate drift level owing to a combination of poor unconfined concrete strength and poor placement of the confinement reinforcement at the toes. However, the performance of the wall up to the failure point was consistent with the predicted behavior based on both the design procedure and the analytical models. **DOI: 10.1061/(ASCE)ST.1943-541X.0000327.** *© 2011 American Society of Civil Engineers.*

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Introduction and Background

As shown in Fig. 1, the hybrid wall configuration investigated in this paper is constructed by stacking rectangular precast concrete wall panels across horizontal joints at the floor levels. A combination of mild (i.e., Grade 400) steel and high-strength unbonded posttensioning (PT) steel is used for lateral resistance. The PT force is provided by multistrand tendons placed inside ungrouted ducts (to prevent bond between the steel and concrete) through the wall panels and the foundation. Thus, the tendons are connected to the structure only at the end anchorages.

Under the application of lateral loads into the nonlinear range, the primary mode of displacement in these walls occurs through gap opening at the horizontal joint between the base panel and the foundation, allowing the wall to undergo large lateral displacements with little damage. Upon unloading, the PT steel provides a restoring force to close this gap, thus reducing the residual (i.e., permanent) lateral displacements of the wall after a large earthquake. The use of unbonded PT tendons delays the yielding of the strands and reduces the tensile stresses transferred to the concrete (thus reducing cracking) as the tendons elongate under lateral loading. Mild steel bars are designed across the horizontal joint at the base to yield in tension and compression and provide energy dissipation through the gap opening/closing behavior of the wall. A predetermined length of these bars is unbonded at the bottom of the base panel (by wrapping the bars with plastic sleeves) to prevent low-cycle fatigue fracture. Both the PT steel and the mild steel contribute to the lateral strength of the wall, resulting in an efficient structure.

Hybrid precast wall structures offer high quality production, relatively simple construction, and excellent seismic characteristics by providing self-centering to the building (i.e., the wall returns to its undisplaced "plumb" position after a large earthquake) and energy dissipation to control the lateral displacements. Despite their desirable characteristics, significant limitations prevent the use of these walls in seismic regions of the United States. Most importantly, Chapter 21 of ACI 318 (2008) specifies that "a reinforced concrete structural system not satisfying the requirements of this chapter shall be permitted if it is demonstrated by experimental evidence and analysis that the proposed system will have strength and toughness equal to or exceeding those provided by a comparable monolithic reinforced concrete structure satisfying this chapter." The hybrid wall system investigated in this paper does not fall within this "emulative" category, thus requiring experimental validation and analysis before its use in practice.

To date, limited tests are available for hybrid precast walls with combined mild steel and PT steel reinforcement for lateral resistance across horizontal joints (Rahman and Restrepo 2000; Holden et al. 2001). These tests are for wall configurations using a single panel (i.e., single horizontal joint at the base). This paper expands the use of hybrid walls to configurations with

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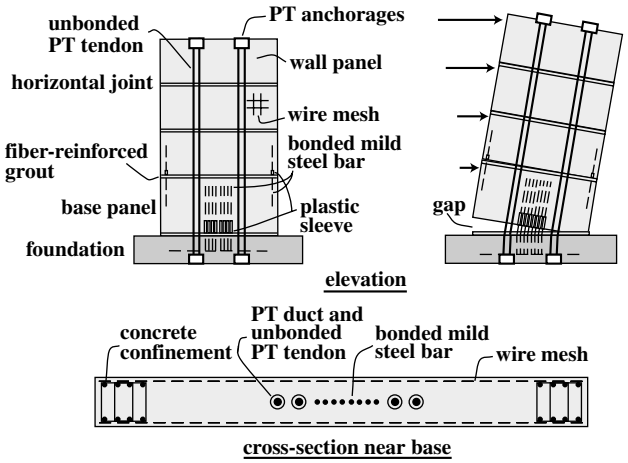

Fig. 1. Elevation, exaggerated displaced position, and cross section of hybrid wall system

multiple horizontal joints. The specific project objectives are to develop: (1) a validated seismic design procedure for the new system; (2) validated analytical models and design tools; and (3) practical guidelines and experimental evidence demonstrating the performance of these structures under lateral loading.

Performance and Design Requirements

The road map for the performance assessment of hybrid precast concrete walls is provided by ACI ITG-5.1 (2007), which lays out the minimum experimental evidence needed for the classification of unbonded posttensioned walls as "special" RC shear walls based on ACI 318 (2008). Among the subjects covered in ACI ITG-5.1 are requirements for the tested wall roof drift, Δ_w, measured to predicted wall lateral strength ratio, PT strand stresses and strains, amount of energy dissipation, wall strength degradation, and shear slip at the horizontal joints. The design is conducted at three wall drift levels as follows: (1) the "linear-elastic" drift, Δ_{we}, which is determined according to ASCE 7 (2005); (2) the "design-level" drift, Δ_{wd}, also determined according to ASCE 7; and (3) the "validation-level" drift, which is prescribed by ACI ITG-5.1 as

$$\Delta_{wm} = 0.9\% \leq (h_w/l_w)0.8\% + 0.5\% \leq 3.0\% \qquad (1)$$

where h_w = height to the top of the wall and l_w = length of the wall. The wall drift, Δ_w, is defined as the lateral displacement at the top of the wall divided by the wall height. Before the testing of the wall, ACI ITG-5.1 requires that a pretest design and analysis procedure for the specimen be established. A few other key ACI ITG-5.1 requirements are the following: (1) the use of a minimum of two wall panels in the test specimen (to model a representative panel-to-panel joint as well as the base-panel-to-foundation joint) unless the full-scale prototype structure uses a single panel for the entire height of the wall; (2) a minimum specimen length scale of one-third; (3) a minimum wall height-to-length aspect ratio of 0.5; and (4) the use of similar reinforcement details and representative building materials as in the full-scale structure.

Overview of Wall Test Specimen, Test Setup, and Procedure

The properties of the wall test specimen described in this paper were determined based on a prototype four-story, regularly shaped precast concrete parking garage [Fig. 2(a)] with an approximate footprint area of 3,770 m^2 (40,600 ft^2). A photograph of the specimen and a schematic of the test setup are shown in Figs. 2(b) and 2(c), respectively. The test was conducted at 0.4-scale, which satisfies the minimum scaling limit of ACI ITG-5.1 (2007). The wall specimen featured two panels: the base panel representing the 1st story and the upper panel representing the 2–4th stories of the prototype wall, thereby satisfying the ACI ITG-5.1 requirement for testing multipanel walls. It was possible to test the upper story panels of the 4-story prototype wall as a single panel because the joints between these panels were designed not to have any gap opening. The lateral load was applied at the resultant location of the 1st mode inertial forces [3.66 m (12 ft) from the wall base], resulting in a base moment to shear ratio of $M_b/V_b = 1.5l_w$. A reversed-cyclic lateral displacement history was used, with three fully repeated cycles at each increment. In addition, a downward axial load of approximately 325 kN (73 kips) was applied at the center of the top of the wall to simulate the service-level tributary gravity loads acting on the prototype structure.

The 0.40-scale wall length, l_w, was 243 cm (96 in.), base panel height, h_{pb}, was 145 cm (57.5 in.), and wall thickness, t_w, was 15.9 cm (6.25 in.). The PT steel consisted of two tendons located 22.9 cm (9 in.) north and south from the wall centerline. The tendons were placed near the wall centerline to reduce the strand strains and also keep the PT ducts away from the critical base panel ends. Each tendon contained three 1.3-cm (0.5-in.) diameter strands [design ultimate strength, $f_{pu} = 1{,}862$ MPa (270 ksi)] with an unbonded length from the top of the wall to the bottom of the foundation beam of approximately 5.48 m (18 ft). The average initial tendon stress, calculated from the measured individual strand forces before the application of the lateral load, was $f_{pi} = 0.55f_{pu}$. The mild steel (i.e., energy dissipating steel) crossing the base joint consisted of four 19-mm-diameter (U.S. No. 6) bars [measured yield strength, $f_{sy} = 448$ MPa (65 ksi)], with one pair of bars placed 15.2 cm (6 in.) north and south from the wall centerline and the other pair 7.6 cm (3 in.) north and south from the centerline. The energy dissipating bars were unbonded over a length of 25.4 cm (10 in.) at the bottom of the base panel. Across the panel-to-panel joint, only two 19-mm-diameter bars were used, with one bar 10.2 cm (4 in.) from each end of the wall. This reinforcement was designed not to yield and therefore, limit any gap opening along the panel-to-panel joint. To prevent strain concentrations in the panel-to-panel joint reinforcement, a short 7.6-cm (3-in.) length of the bars was unbonded at the bottom of the upper panel. The design unconfined concrete strength for the wall was 41 MPa (6.0 ksi) and the design confined concrete strength (at the toes of the base panel) was 62 MPa (9.0 ksi). However, the 141-day unconfined concrete strength for the base panel was only 33 MPa (4.8 ksi) on the day that the wall was tested. At the base and panel-to-panel joints, fiber-reinforced grout (polypropylene microfilament fibers at 0.065% by volume) with a 7-day compressive strength of 31 MPa (4.5 ksi) was used. This grout was deliberately designed to have a smaller strength than the design strength of the unconfined wall panel concrete to provide a relatively soft contact surface at the base.

Prototype Wall Design

Specific guidelines for the design of special unbonded posttensioned precast shear walls satisfying ACI ITG-5.1 (2007) are given in ACI ITG-5.2 (2009). The prototype parking garage that formed the basis of the wall test specimen was designed in collaboration with the Consulting Engineers Group (CEG), Texas, following the

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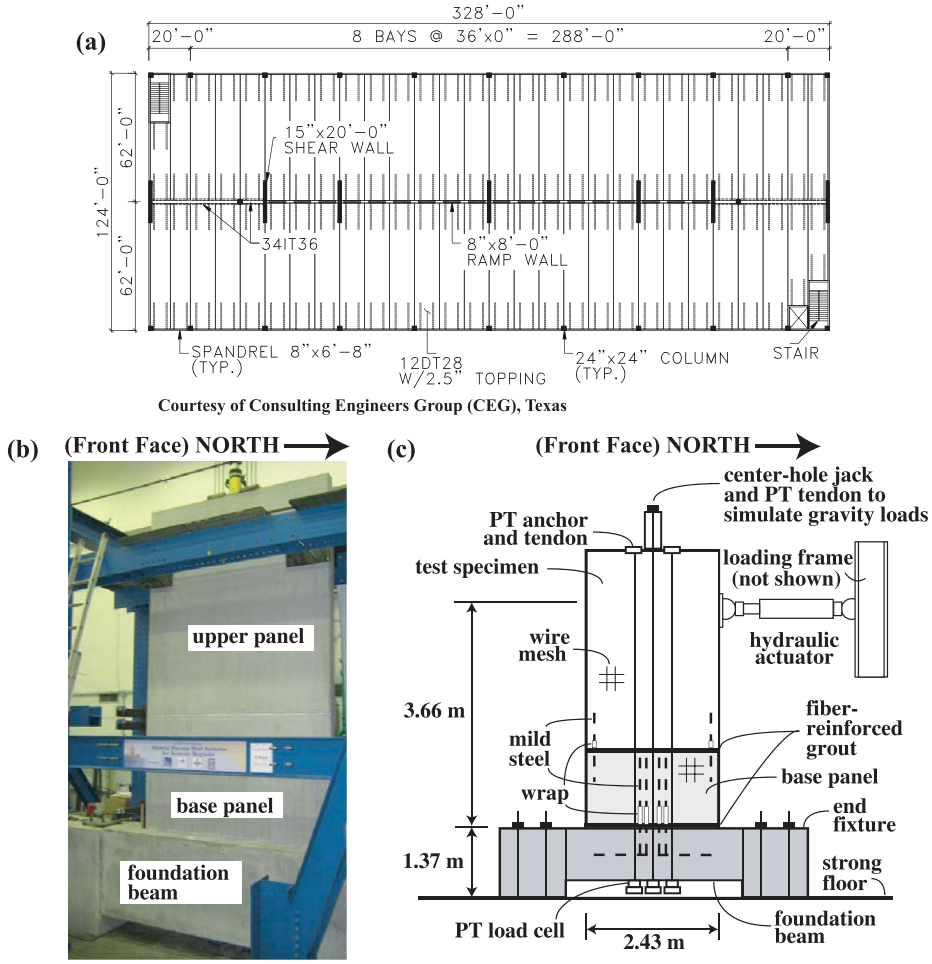

Fig. 2. Test overview: (a) prototype building plan (image courtesy of Consulting Engineers Group, Texas); (b) photograph of specimen (image courtesy of the authors); and (c) test setup

basic guidelines in ACI ITG-5.2 and those in ACI 318 (2008). As shown in Fig. 2(a), the lateral load resistance of the building in the N–S direction is provided by seven hybrid shear walls. One of the objectives of the research was to utilize conventional methods for the seismic design of the wall. In accordance with this objective, the equivalent lateral force procedure in ASCE 7 (2005) was used to determine the full-scale base moment demand for the exterior walls (where the lateral force demand is largest considering accidental torsion effects) as $M_{we} = 24{,}950$ kN-m (18,400 kip-ft). The structure was designed for a site in Los Angeles, California, where the seismic response coefficient for the building was calculated as $C_s = 0.167$ g. A response modification factor of $R = 6.0$ for special RC shear walls was used. For the selected full-scale wall dimensions of $h_w = 13.7$ m [45 ft, with 3.66 m (12 ft) for the 1st story height and 3.35 m (11 ft) for the upper stories] and $l_w = 6.1$ m (20 ft), resulting in an aspect ratio of 2.25, the validation-level drift from Eq. (1) (i.e., the drift level at which validation is sought according to ACI ITG-5.1) is $\Delta_{wm} = 2.30\%$ and the design-level drift from ASCE 7 is $\Delta_{wd} = 0.65\%$. In the calculation of Δ_{wd}, a displacement amplification factor of $C_d = 5.0$ was applied to the linear-elastic drift of the wall, $\Delta_{we} = 0.13\%$ (flexural plus shear displacements corresponding to M_{we}), which was determined by using an assumed shear area of $A_{sh} = 0.67A_{\text{gross}}$ and an assumed moment of inertia of approximately $I_r = 0.5I_{\text{gross}}$ (with the 0.5 factor representing the reduction in the flexural stiffness of the wall due to gap opening over approximately 20% of the wall length at the base). When using the measured concrete strength (as opposed to the design strength) in the drift calculations, the linear-elastic and design-level drifts become $\Delta_{we} = 0.14\%$ and $\Delta_{wd} = 0.70\%$, respectively.

Shear slip across the horizontal joints was prevented following the design guidelines in ACI ITG-5.2 (2009). The nominal shear slip strength at the base joint was taken as the coefficient of friction, μ (assumed as 0.5), times the concrete compression stress resultant, C, at the interface. The design shear slip strength was calculated as the nominal strength times a strength reduction factor, φ, of 0.75. For axial-flexural behavior, the test specimen was intentionally designed not to have any significant overcapacity (neither in force nor displacement) beyond the minimum requirements of ACI ITG-5.2 and ACI 318 (2008). This decision was made in an effort to determine if the procedure and requirements used in the design of the wall were overly conservative, not conservative enough, or reasonable. The design procedure, described in Smith and Kurama (2009), provided specific steps to determine the PT and mild steel areas and strains, the confined concrete detailing at the wall toes, and the behavior at the base-panel-to-foundation and panel-to-panel joints (only the base joint was allowed to have significant gap opening). Although other details such as external armor plates may be possible, conventional closed steel hoops were used as concrete confinement at the wall toes to minimize the differences of the test specimen from typical construction. The energy dissipating mild steel reinforcement was designed to resist approximately 34% of the total wall base moment demand, M_{we}, with the remaining base moment demand resisted by the PT steel and the applied gravity

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load. According to ACI ITG-5.2, this design represents the approximate upper bound for the amount of mild steel that can be placed across the base joint while maintaining the self-centering capability of the structure.

Analytical Models

Two different analytical models were developed for the test structure: a fiber-element model and a finite-element model. As described in Kurama (2002) and shown in Fig. 3(a), the fiber-element model used a modified version of the DRAIN-2DX Program (Prakash et al. 1993) as the analysis platform, with fiber beam-column elements to represent the precast wall panels and truss elements for the unbonded PT steel. The unbonded portions of the mild steel bars crossing the base joint were also modeled as truss elements outside the fiber cross sections (to capture the uniform distribution of strains over the unbonded length), but the bars across the upper joint were modeled as part of the fiber elements because no significant gap opening was expected at this joint. According to ACI ITG-5.2 (2009), additional "debonding" of the energy dissipating bars occurs over a length of 2 times the bar diameter, d_b, under reversed-cyclic loading up to the validation-level drift. For modeling purposes, this debonding length was assumed to remain constant throughout the analysis. The grout pads at the horizontal joints were not modeled (i.e., the grout was assumed to behave the same as the wall concrete). Gap opening along the base joint was modeled by setting the tension strength of the concrete fibers at the bottom of the base panel to zero. Different from the pretest model in Smith and Kurama (2009), measured material and geometric properties were used in the analysis of the structure in this paper. Small modifications were also made to the initial stiffness of the PT steel to account for the seating of the anchors and the local concrete deformations around the anchors. Additionally, as shown in Fig. 3(b), the cyclic stress-strain behavior of the energy dissipating bars was slightly modified from the measured monotonic behavior (e.g., the yield plateau was not modeled).

To demonstrate that the general behavior and some of the unique characteristics of hybrid walls can be captured with more common analysis techniques, a basic finite-element (FE) model for monotonic lateral load analysis was created in ABAQUS [2009; Fig. 3(c)]. This model intentionally incorporated several simplifying assumptions appropriate for the design office, such as elastoplastic concrete behavior (by using the measured strength of the unconfined wall concrete as the yield point), elastoplastic mild steel behavior (by using the measured yield strength), "hard" contact at the base-panel-to-foundation joint, and lumped PT and mild steel areas at the wall centerline. The assumed elastoplastic concrete behavior prevented the FE model from capturing concrete cracking; however, gap opening and contact behavior at the base joint were modeled.

Measurement Systems

Two different measurement systems were used to monitor the test structure: (1) a conventional system featuring displacement transducers, rotation transducers, load cells, and strain gauges; and (2) a three-dimensional digital image correlation (3D-DIC) system that uses a noncontact optical technique. As partially shown in Fig. 4(a), the conventional system included 57 channels of data with 23 displacement transducers, 5 rotation transducers, 9 load cells, and 20 strain gauges. The displacement transducers, consisting of string potentiometers and linear variable differential transformers (LVDTs), were used to measure the in-plane displacements of the wall panels and foundation beam, gap/contact behavior and horizontal slip across the base-panel-to-foundation and panel-to-panel joints, shear deformations of the base panel, and extent of the concrete crushing zone at the wall toes. The rotation transducers were used to record the in-plane rotations of the base panel, upper panel, and foundation beam. The load cells were used to measure the applied lateral force, the vertical force simulating service-level gravity loads on the wall, and the forces in the individual PT strands. Finally, the strain gauges were used to measure the strains in the mild steel bars across the base panel and upper panel joints, compression strains of bars embedded inside the confined concrete at the wall toes, and strains in the horizontal edge reinforcement at the bottom of the base panel.

As shown in Fig. 4(b), the 3D-DIC system consisted of two cameras measuring the in-plane and out-of-plane displacements of the back face (i.e., west surface) of the base panel over a region of approximately one half the panel length and one half the panel height from the south toe [white-painted region in Fig. 4(a)]. A portion of the foundation beam was also included in the "field of view," providing unprecedented information on the response across the base joint. 3D-DIC has been previously used to monitor deformations in civil engineering applications (e.g., Orteu 2009; McGinnis et al. 2005; Corr et al. 2007). In this method, a stochastic pattern is applied to the surface of the monitored object, and stereo pairs of photographs of this pattern are captured before and after load events. The captured digital images are divided into regions that are several pixels across, called facets. Then, these facets are

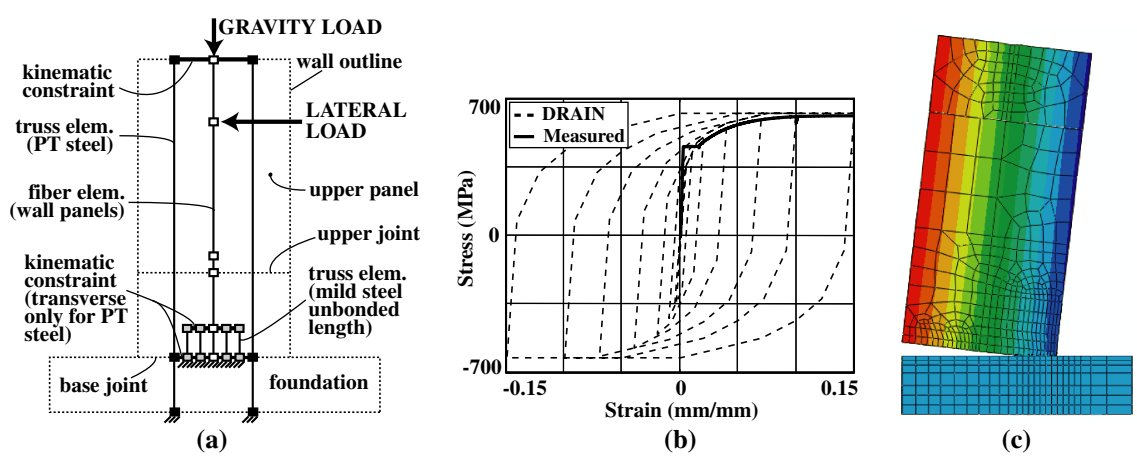

Fig. 3. Analytical modeling: (a) fiber-element (DRAIN) model; (b) mild steel stress-strain behavior; and (c) finite-element (ABAQUS) model

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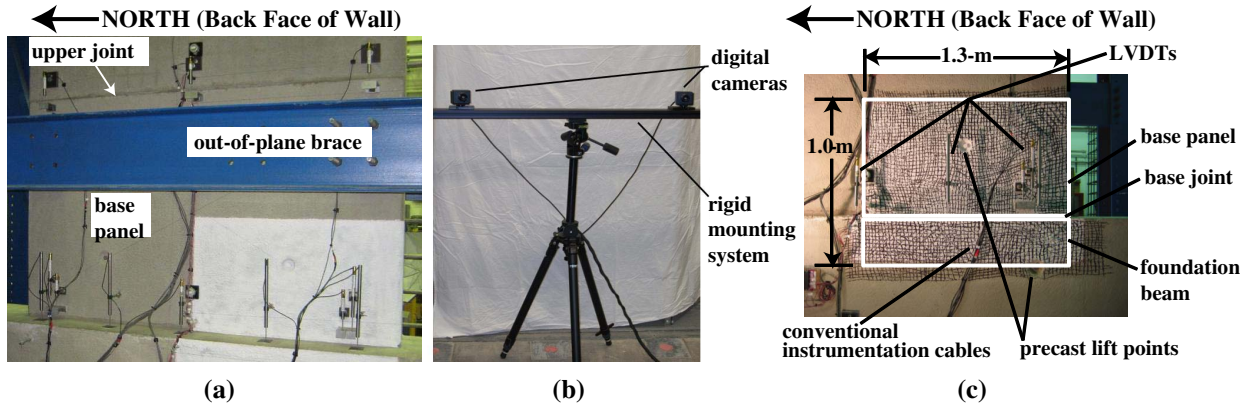

Fig. 4. Measurement systems: (a) conventional sensors; (b) 3D-DIC camera setup; and (c) 3D-DIC field of view (images courtesy of the authors)

tracked through an image series by using pattern recognition and photogrammetric triangulation principles, thus yielding essentially full-field, 3D surface displacements and strains of the view area. The accuracy of 3D-DIC scales somewhat linearly with the field of view (FOV). For the FOV used in this study [approximately 1.3 m (4.3 ft) horizontal by 1.0 m (3.3 ft) vertical, including both the base panel and foundation regions; see Fig. 4(c)], the in-plane accuracy for displacements is conservatively stated as ± 30 micrometers. Advantages of the technique include ease of setup and the full-field, 3D results. Disadvantages include inability to capture deformations after cover spalling or inside the specimen.

Measured, Design, and Analytical Behaviors

Fig. 5(a) depicts the measured base shear force, V_b versus wall drift, Δ_w behavior of the test specimen and Fig. 5(b) shows the corresponding predicted behaviors from the fiber-element and finite-element models described previously. The wall drift, Δ_w, was measured as the relative lateral displacement of the wall between the lateral load location and the foundation divided by the height to the lateral load. The wall specimen was loaded in a slightly unsymmetrical manner due to the movement of the foundation beam during the test; however, all data presented in this paper have been corrected to isolate the wall response from this foundation movement.

Fig. 5(c) shows the lower half of the wall during the third cycle at $\Delta_w = +1.90\%$ (note the gap opening at the north end). The specimen sustained three cycles at a maximum positive drift (with the wall displaced southward) of $\Delta_w = +1.90\%$ and a maximum negative drift of $\Delta_w = -1.55\%$ before failure due to the crushing of the confined concrete at the toes. Fig. 5(d) shows the south toe of the wall at the end of the test. The first confinement hoop was placed at a significant angle with the horizontal, resulting in a large

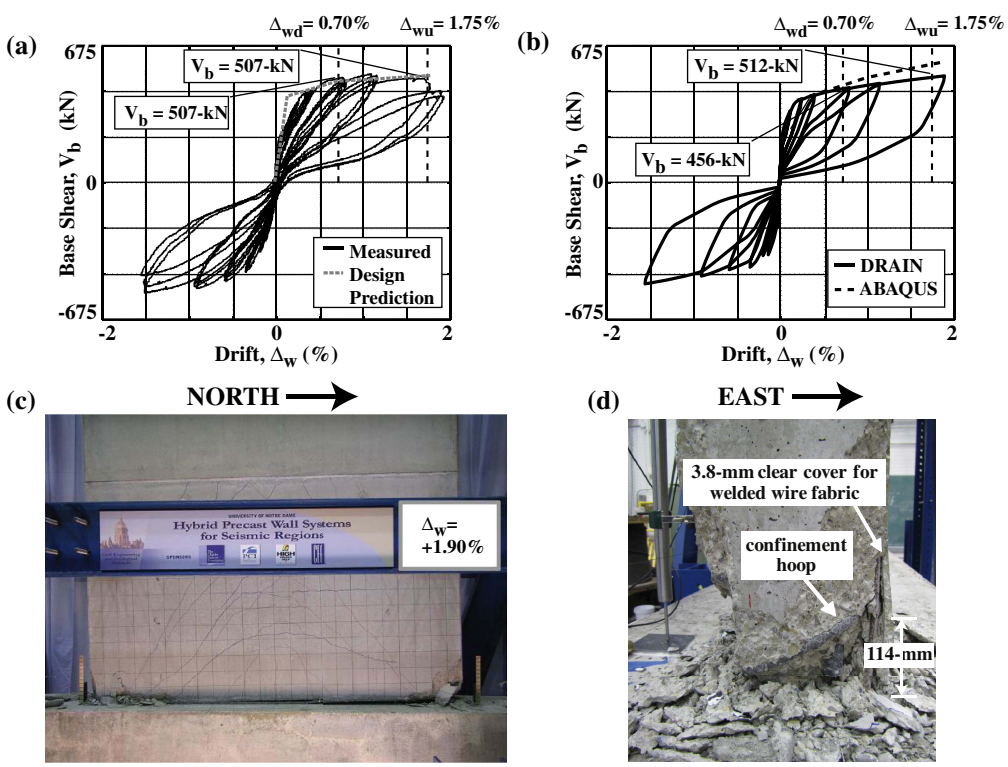

Fig. 5. Wall behavior: (a) measured V_b-Δ_w response; (b) analytical V_b-Δ_w response; (c) damage at third cycle of $\Delta_w = +1.90\%$; and (d) south toe of wall after completion of test (images courtesy of the authors)

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Table 1. Comparisons of Design Predictions and Measured Results

	Linear-elastic drift, $\Delta_{we} = +0.14\%$		Design-level drift, $\Delta_{wd} = +0.70\%$		Failure-level drift, $\Delta_{wu} = +1.75\%$	
Design parameter	Predicted	Measured	Predicted	Measured	Predicted	Measured
Applied lateral load	423 kN	311 kN	493 kN	507 kN	525 kN	507 kN
Intermediate mild steel strains	0.41 and 0.49%	0.22 and 0.35%	2.2 and 2.6%	1.1 and 1.6%	5.5 and 6.5%	—
Average PT steel stresses	0.55 and $0.57f_{pu}$	0.55 and $0.57f_{pu}$	0.62 and $0.69f_{pu}$	0.63 and $0.68f_{pu}$	0.73 and $0.84f_{pu}$	0.70 and $0.82f_{pu}$
Maximum gap opening	2.8 mm	1.3 mm	15 mm	15 mm	37 mm	39 mm
Neutral axis depth at base	406 mm	488 mm	340 mm	276 mm	351 mm	467 mm

region of unconfined concrete at the bottom of the base panel [at the east face of the panel, the first hoop was located 11.4 cm (4.5 in.) from the bottom rather than the design location of 5.0 cm (2 in.)]. Although not as extreme as the south toe, the hoop placement at the north toe was also misaligned. This misalignment of the confinement hoops, combined with the low unconfined concrete strength [33 MPa (4.8 ksi)] of the critical base panel, resulted in the failure of the wall at a lower drift than it was designed for.

Although neither analytical model was able to capture the strength degradation that occurred during the large displacements of the wall, the analytical results in Fig. 5(b) show similar behaviors as the measured behavior in Fig. 5(a). Comparisons between the measured results and the corresponding design predictions for key aspects of the test structure are provided in Table 1, with the trilinear design prediction for the V_b-Δ_w behavior designated by the light dashed lines in Fig. 5(a). This trilinear V_b-Δ_w behavior consists of the design predictions at the linear-elastic wall drift, $\Delta_{we} = 0.14\%$, the design-level drift, $\Delta_{wd} = 0.70\%$, and the "failure-level" drift, $\Delta_{wu} = 1.75\%$. Instead of the validation-level drift, $\Delta_{wm} = 2.30\%$, which was not achieved by the specimen, the design comparisons are provided by using the failure-level drift, Δ_{wu}, which was determined as the measured drift when confined concrete crushing was first observed, and for the subsequent drift cycles, significant strength degradation occurred in the V_b-Δ_w behavior. In general, with the exception of the energy dissipating, mild steel strains and the neutral axis depth at the base (as described subsequently), the design predictions at Δ_{wd} and Δ_{wu} compare well with the measured response of the wall. Because of the linear approximation, the design predictions at Δ_{we} are different than the measured values; however, the calculated shear force at Δ_{we} (423 kN) provides a good estimate for the point of significant stiffness reduction in the structure while also providing a reasonable approximation to the initial stiffness of the wall.

Progression of Damage

Fig. 6 shows front face photographs of the south end of the wall at $\Delta_w = +0.25\%$ (third cycle), $+0.40\%$ (first cycle), and $+0.80\%$ (first cycle) along with the corresponding vertical (i.e., axial) strains from the 3D-DIC data measured at the back face. Data drops in the strain plots (i.e., white regions of the plots) are locations where conventional sensors were mounted to the wall, obstructing the field of view of the 3D-DIC cameras. Compression strains of $-1{,}525$ microstrain are evident at the toe of the wall at $\Delta_w = +0.25\%$. Clearly visible in these images are increased axial strains ($-1{,}960$ microstrain; a value associated with yield in a Grade 400 rebar, for example) and diagonal cracks terminating in the plastic hinge region of the base panel as the wall is pushed to $\Delta_w = +0.80\%$. As shown by both the photographs and the strain plots, cracking initiated during the first cycle to $\Delta_w = +0.40\%$. The 3D-DIC system was able to capture the initiation and location of the cracks with excellent accuracy. Note that the dark shaded regions in the strain plots are not indicative of the crack widths, which remained small throughout the test (the cracks visible in the photographs were highlighted with markers during the test for enhanced viewing). Furthermore, any small differences between the crack patterns in the photographs and those in the strain plots can be explained by differences in the crack locations between the front and back faces of the wall.

The initiation of cover concrete spalling occurred during the 0.80% drift cycles [see the south toe of the wall in Fig. 6(c)], after the design-level drift, Δ_{wd}, was exceeded. Significant crushing of the confined concrete at the wall toes was not present until approximately $\Delta_w = +1.75\%$ (final drift series), after which strength degradation was evident as reflected by the V_b-Δ_w behavior during the second and third cycles at $\Delta_w = +1.90\%$ in Fig. 5(a). Although the design clear cover was 1.3 cm (0.50 in.) for the welded wire fabric placed on each face of the wall, the actual clear cover was as little as 0.38 cm (0.15 in.) near the south end of the base panel. As shown in Fig. 5(d), this resulted in the delamination of the wire fabric near the end of the test. The test was stopped after the third cycle in this displacement series because the total strength degradation of the wall was approximately 20%, which is the strength degradation limit given by ACI ITG-5.1 (2007). Because of the smaller actual displacements reached, the performance of the wall in the negative drift direction was somewhat better, with a slightly smaller amount of strength degradation (~16%). No concrete cracking or spalling was observed in the upper panel throughout the test [see Fig. 5(c)].

A useful way to look at Fig. 6 is to plot the extent of significant compression strains at the wall toes when the peak drift is reached during each displacement cycle. For example, the solid line in Fig. 7(a) shows the height from the top of the foundation [at approximately 2.5–5 cm (1–2 in.) in the horizontal direction from the south end of the wall] where the 3D-DIC axial compression strains in the base panel exceeded $-1{,}000$ microstrain. The dashed line depicts the height of concrete spalling at the same location. It is likely that concrete compression strains between these two lines are indicative of plastic hinge formation. Similarly, Fig. 7(b) depicts the average axial compression strains determined by using three LVDTs [with gauge lengths of 9, 18, and 27 cm (3.5, 7, and 10.5 in.) measured from the top of the foundation beam], placed approximately 17.5–22.5 cm (6.8–8.8 in.) in the horizontal direction from the south end of the wall. Most of the compression deformations of the wall occurred within the 9-cm (3.5-in.) gauge length. It can be deduced that the LVDT strains in Fig. 7(b) are smaller than the 3D-DIC strains in Fig. 7(a), which is expected because the LVDT strains are average strains measured at a larger distance from the wall end (because of data drops, 3D-DIC measurements could not be made at the LVDT locations).

Energy Dissipating Mild Steel Behavior

Because the mild steel bars crossing the base joint serve as the main energy dissipater for the wall, it is essential for these bars to yield well before the design-level drift but not fracture before the validation-level drift. Fig. 8(a) shows the four mild steel bars at the bottom of the base panel before concrete placement. The 25.4-cm (10-in.) long plastic-wrapped unbonded length of the bars

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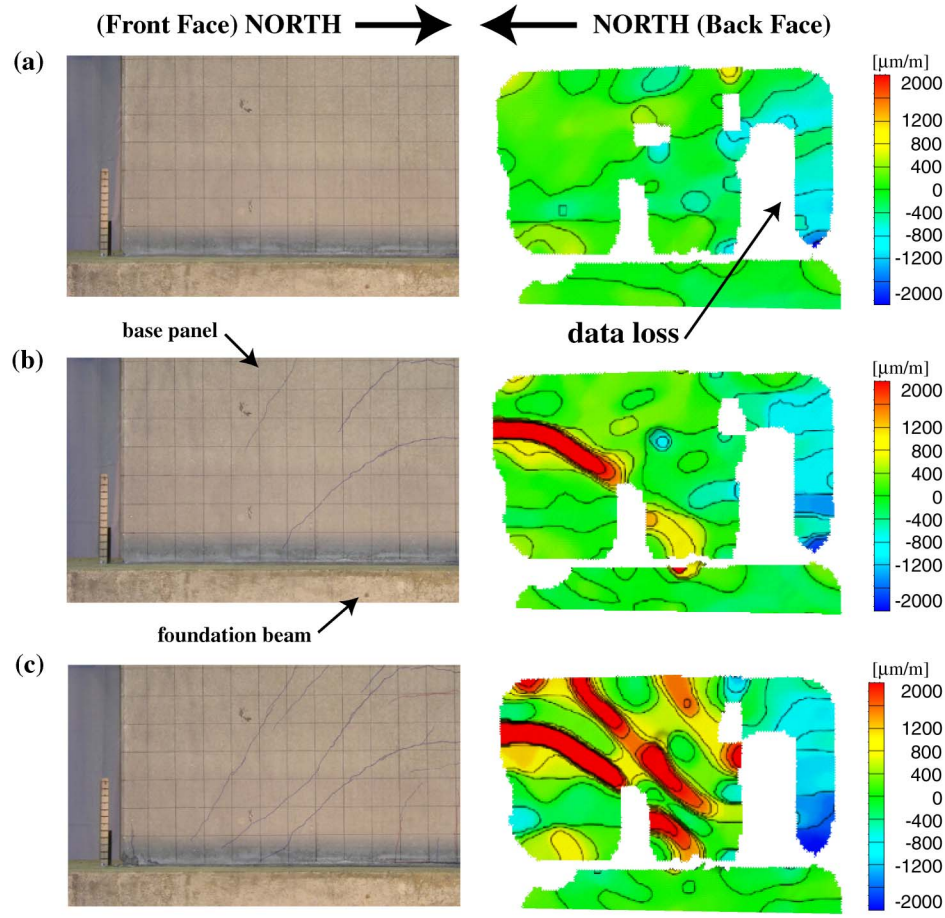

Fig. 6. Progression of damage: (a) $\Delta_w = +0.25\%$ third cycle; (b) $\Delta_w = +0.40\%$ first cycle; and (c) $\Delta_w = +0.80\%$ first cycle

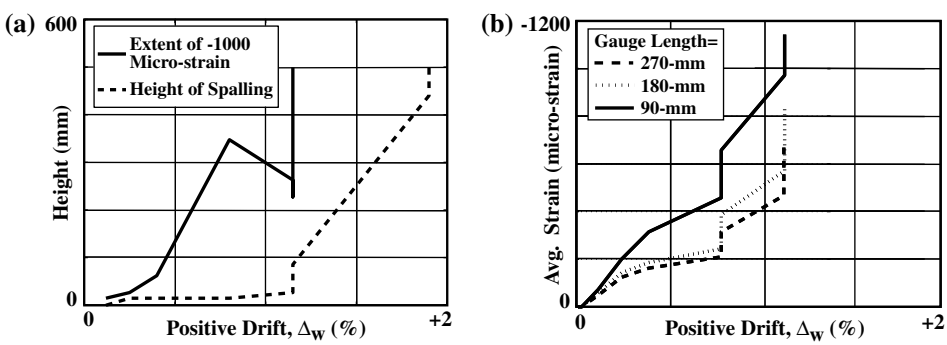

Fig. 7. Axial compression deformations at south toe: (a) significant axial strains and spalling; and (b) average LVDT strains

can be seen in the photograph, which was done to reduce the steel strains and thus, prevent low-cycle fatigue fracture as the wall was displaced. Additionally, the bars were located near the wall centerline to further reduce the strains and, in turn, reduce the required unbonded length.

Fig. 8(b) compares the measured strains (using strain gauges placed within the unbonded length) and the fiber-element analytical model strains of the middle two bars (referred to as the north and south intermediate bars) located ± 7.6 cm (3 in.) from the wall centerline. The strains in the mild steel bar lumped at the centerline of the ABAQUS wall model are also shown. As designed, the mild steel bars yielded (measured yield strain, $\varepsilon_{sy} = 0.00227$ mm/mm) relatively early in the test and before the design-level drift, $\Delta_{wd} = 0.70\%$. As shown in Table 1, the measured strains at $\Delta_{wd} = +0.70\%$ (0.011 and 0.016 mm/mm for the south and north bars, respectively) are significantly smaller than the corresponding design predictions (0.022 and 0.026 mm/mm). The differences in the north and south bar stains are attributable to the different elongations of the two bars when the wall was displaced in one direction. At the failure-level drift of $\Delta_{wu} = +1.75\%$, the predicted strains for the south and north intermediate bars were 0.055 and 0.065 mm/mm, respectively (no measurements were possible at this drift level due to gauge failure). The overprediction of the steel strains may have occurred because of potential slip of the bars from the foundation beam; however, it was not possible to visually observe this slip. The differences in the measured, design, and analytical steel strains may also be related to differences in the neutral axis depth of the wall cross section at the base (described

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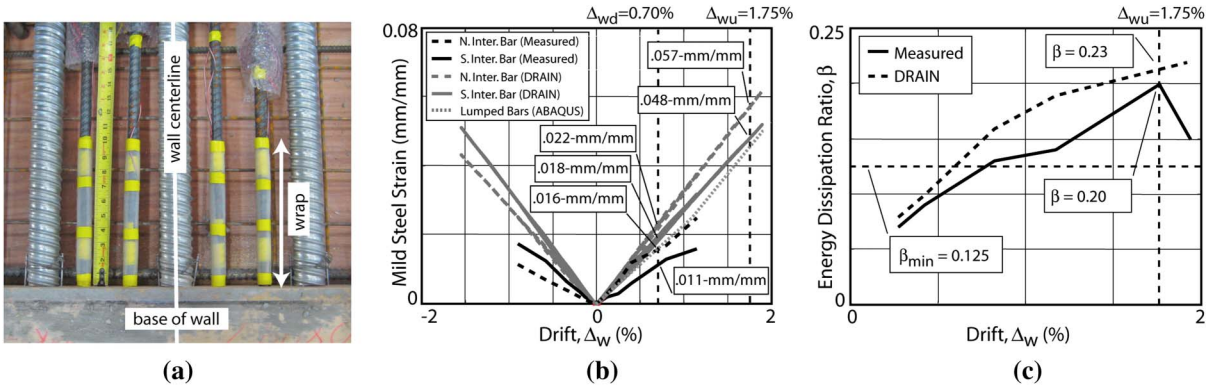

Fig. 8. Energy dissipation: (a) mild steel bar placement across base joint (image courtesy of the authors): (b) measured and analytical mild steel strains; and (c) energy dissipation ratio

subsequently) and differences in the assumed additional debonding of the bars under reversed-cyclic loading. As stated previously, a debonding length of $2d_b$ was used for the design of the wall at the validation-level drift, but for modeling purposes, this debonding length was assumed to remain constant during the entire loading history. Concrete cores taken from the base panel through the energy dissipating bars after the test supported the additional debonding length of $2d_b$ at failure.

To quantify the energy dissipation of the structure, ACI ITG-5.1 (2007) uses the energy dissipation ratio β, which is defined as "the ratio of the measured energy dissipated by the test module during reversing cyclic displacements between given measured drift angles to the maximum theoretical energy that can be dissipated for the same drift angles." ACI ITG-5.1 requires that β be not less than 0.125 at the validation-level drift. The solid line in Fig. 8(c) shows the measured β of the test structure plotted against the wall drift. The third cycle for each drift level was used to calculate β, except for the last series in which both the first and third cycles were used to assess the effect of strength degradation on the energy dissipation of the wall at the end of the test. The specimen exceeded the minimum β requirement at drift levels greater than $\Delta_w = 0.80\%$ and achieved a maximum β of 0.20 at $\Delta_{wu} = 1.75\%$. The corresponding "equivalent viscous damping ratio," ξ_{eh} (Kurama 2001, 2005) for a linear-elastic system with the same natural frequency (i.e., ignoring frequency shift) can be estimated as $\xi_{eh} \approx \beta(2/\pi)$, resulting in $\xi_{eh} \approx 0.13$ at $\Delta_{wu} = 1.75\%$. The strength degradation during the last drift series resulted in a reduction in β. The energy dissipation from the fiber-element model [dashed line in Fig. 8(c)] tended to overestimate the measured energy dissipation, especially during intermediate drift levels. This is likely a consequence of the inadequacy of the model to accurately match the maximum nonlinear strains and behavior of the mild steel bars during reversed-cyclic loading.

PT Steel Behavior

The PT steel provides the main restoring force for the wall, allowing the structure to return to its initial undisplaced position upon removal of the lateral loads. Fig. 9(a) depicts this restoring force by plotting the normalized average stress in each of the north and south tendons [calculated as the sum of the measured strand forces divided by $A_p f_{pu}$, where A_p is the total area of the three strands in each tendon and $f_{pu} = 1{,}862$ MPa (270 ksi)]. As shown in Fig. 9(b) and Table 1, the tendon stresses from the analytical models and the design predictions agree with the measured data. Similar to the mild steel reinforcement, the differences in the north and south tendon stresses are attributable to the different elongations of the north and south tendons as the wall was displaced laterally (the ABAQUS results do not show this difference because the entire PT steel area was lumped at the wall centerline). Consistent with the design expectations, the PT tendons remained essentially linear-elastic throughout the test, which was possible because the strands were unbonded over their length. Losses in the measured PT steel stresses can be seen during the second and third cycles to $\Delta_w = +1.90\%$ in Fig. 9(a). These losses, which were not captured by the analytical models, occurred due to the crushing of the concrete at the wall base (which resulted in a small amount of axial "shortening" of the wall as shown subsequently), and not because of the nonlinear straining of the strands.

Gap Opening and Neutral Axis Depth

Consistent with the expected behavior of the wall, the structure opened a significant gap at the base-panel-to-foundation joint. As designed, the gap opening at the upper panel-to-panel joint was negligible. Fig. 10(a) shows the vertical (upward positive) 3D-DIC displacements measured near the base of the wall at the first cycle to $\Delta_w = +0.80\%$. Displacements associated with rigid body rotation dominate the results, which is consistent with the small shear distortions measured during the test as described later. Data reduction by comparing the vertical displacements of adjacent points on the foundation beam and the wall base allowed the gap opening displacements to be determined. For example, Fig. 10(b) shows the maximum gap opening displacements across the joint (i.e., vertical size of gap) at the extreme north and south ends of the wall as determined by 3D-DIC and by using five LVDTs plus a rotation transducer placed at the wall base. The two measurement systems exhibit excellent agreement. Because only the south end of the specimen was monitored, the 3D-DIC gap opening displacements at the north end were determined by extrapolating the data along the wall length (the linearity of the displacements along the wall allowed for this extrapolation). At the failure-level drift of $\Delta_{wu} = +1.75\%$, the gap opening at the north end of the wall was approximately 3.9 cm (1.52 in.) at the base joint and 0.15 cm (0.06 in.) at the upper panel-to-panel joint. The analytical model results, shown in Fig. 10(c), compare well with the measured gap opening for the base joint at both the design-level and failure-level drifts. Although the models cannot capture concrete cracking, they produce a realistic representation of the gap opening and the same rigid body behavior that was measured during the test.

Similarly, using the 3D-DIC data, Fig. 10(d) plots the vertical displacements across the base joint versus distance from the south end of the wall (normalized with l_w) for increasing positive Δ_w (where contact between the base panel and the foundation occurred at the monitored south end). The location where the displacements change sign signifies the location of the neutral axis of the section

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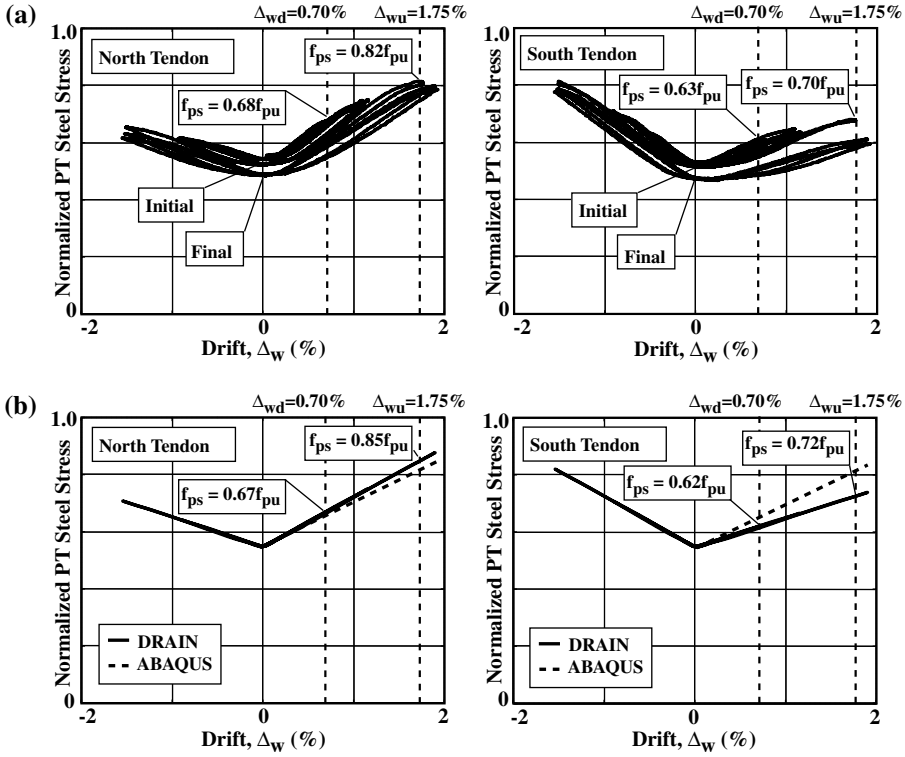

Fig. 9. Average PT steel stresses: (a) measured; and (b) analytical

(i.e., "contact depth" along the joint). Fig. 10(e) plots the corresponding neutral axis depth at the south end of the wall. The results are shown for the first cycle in each drift series, except for the last series in which all three cycles are shown. Referring to Table 1, the design procedure was able to predict the vertical size of the gap opening at the base joint; however, the differences between the measured and design values are greater for the neutral axis depth. The predicted neutral axis depths at the design-level and failure-level drifts were 34.0 cm (13.4 in.) and 35.1 cm (13.8 in.), respectively, with the corresponding measured neutral axis depths of 27.6 cm (10.9 in.) and 46.7 cm (18.4 in.), respectively. The larger discrepancy between the predicted and measured results at the failure-level drift could be related to the premature crushing of the concrete, which, as shown in Figs. 10(d) and 10(e), caused an increase in the neutral axis depth at the wall base.

The vertical displacement of the wall at the top, which is related to the amount of gap opening along the horizontal joints, is another measure that can be used to validate the analytical models. For this purpose, the light solid line in Fig. 10(f) shows the measured vertical displacement at the centerline of the wall at the location of the applied lateral load. The results from the analytical models, shown with the dark dashed and dotted lines, compare well with the test results. A small amount of axial shortening of the wall is evident from the measured data, which occurred as a result of the crushing of the concrete at the wall base during the later cycles. This behavior was not captured by the analytical models.

Shear Deformations

The solid and dashed lines in Fig. 11(a) show the shear distortion angle, γ (calculated as described in Oesterle et al. 1976) of the base panel measured by using two diagonally placed string potentiometers (SP) and 3D-DIC, respectively. The 3D-DIC system measured the deformations over only the bottom south quarter of the base panel, whereas the string potentiometers measured over essentially the entire panel length and height. As expected, the shear distortion is greater over the bottom of the panel compared with the entire panel. When comparing the SP results with the ABAQUS results (dotted lines) for the shear distortion over the entire panel, the measured deformations get noticeably greater beyond $\Delta_w = 0.40\%$. This can be attributed to the increased shear distortions due to the cracking of the base panel, which was not included by the finite-element model.

In general, the shear distortions of the base panel were small, which becomes more obvious when, in Fig. 11(b), the distortion angle of the panel is plotted as a percentage of the total drift angle to the top of the panel, Δ_p (defined as the relative lateral displacement between the top of the base panel and the foundation divided by the height to the top of the base panel). It is concluded that the wall did not undergo significant shear deformations despite the relatively low M_b/V_b ratio of $1.5l_w$. Note that Δ_p is nearly equal to Δ_w because the lateral displacements of the wall were governed by rigid body rotations through the gap opening at the base.

By examining the 3D-DIC data for the relative horizontal displacements between adjacent points on either side of the base-panel-to-foundation joint [e.g., see Fig. 11(c), which shows the horizontal (southward positive) displacements near the base of the wall at the first cycle to $\Delta_w = +0.80\%$], the horizontal slip across the joint can be determined. Fig. 11(d) shows the measured horizontal slip at the centerline of the wall and at the south end. Because of the positioning of the 3D-DIC cameras, it was possible to measure the slip at the wall centerline during both positive and negative drifts, whereas the slip at the south end of the wall was measured during positive drifts only (when the monitored south end was in compression) and the slip at the north end could not be measured. The slip at the base joint was extremely small, with a maximum value of approximately 0.2 cm (0.08 in.) at the peak wall drift of $\Delta_w = +1.90\%$. This amount of slip did not affect the performance of the wall in any undesirable way. From the almost

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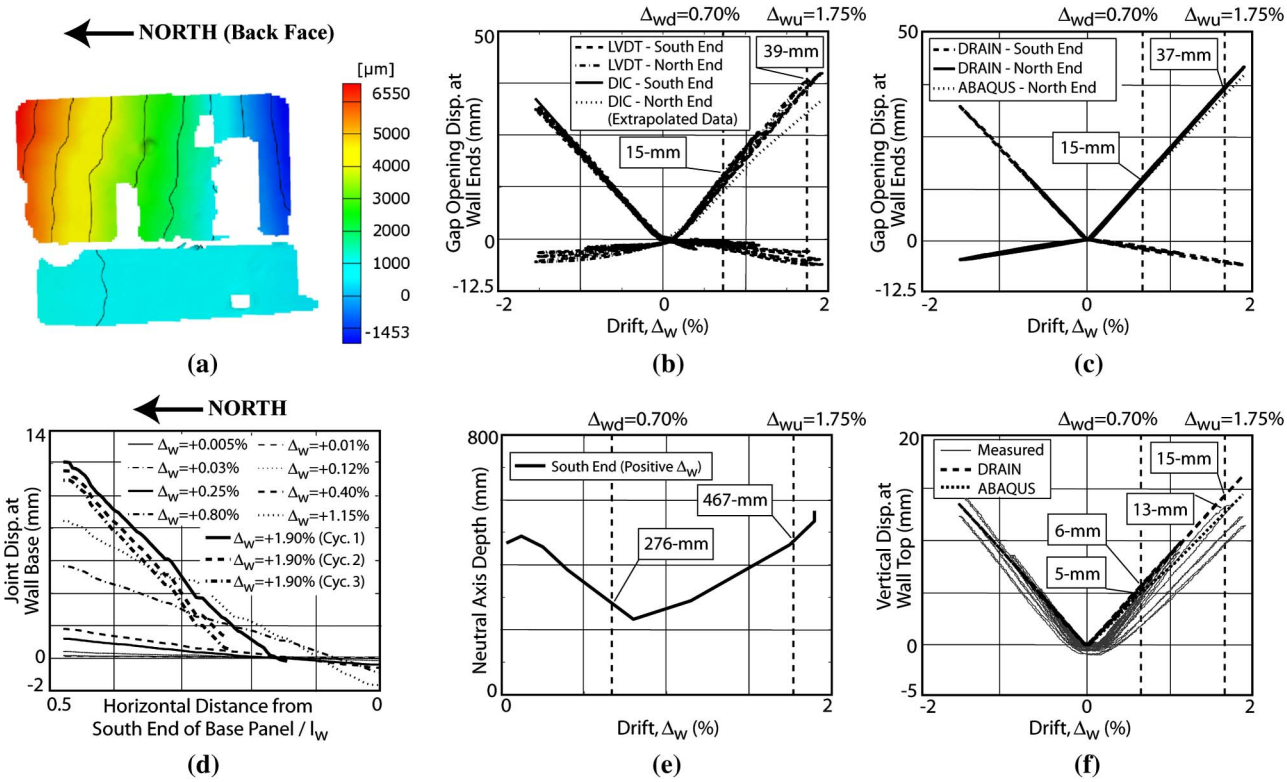

Fig. 10. Gap opening and contact behavior at wall base: (a) vertical displacements at first cycle of $\Delta_w = +0.80\%$; (b) gap opening displacements at wall ends, measured; (c) gap opening displacements at wall ends, analytical; (d) measured joint displacements versus distance from south end; (e) measured neutral axis depth; and (f) vertical displacements of wall at lateral load level

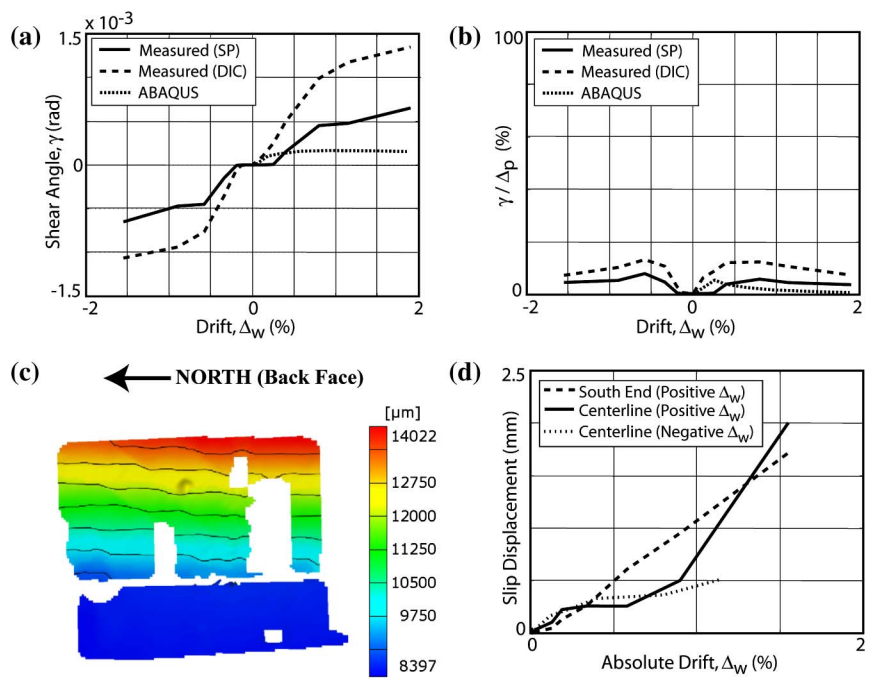

Fig. 11. Shear deformations: (a) shear distortion of base panel; (b) shear distortion normalized by total base panel drift; (c) horizontal displacements of base panel and foundation at first cycle of $\Delta_w = +0.80\%$; and (d) shear slip at base joint

linear relationship between base slip and wall drift in Fig. 11(d), it can be stated that the crushing of the concrete at the wall toes near the end of the test did not result in a disproportional increase in slip. No slip was observed at the upper panel-to-panel joint of the wall.

Summary and Conclusions

This paper presents the measured lateral load behavior of a 0.40-scale hybrid precast concrete wall test specimen in comparison with design and analytical predictions. Overall, the wall system

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performed as designed; however, failure occurred prematurely due to lower than specified concrete strength and poor placement of the confinement hoops at the wall toes. Nevertheless, the design procedure was able to result in a structure that achieved the required lateral strength and energy dissipation as well as the required performance of the mild steel and PT steel reinforcement. The wall behaved essentially as a rigid body dominated by gap opening at the horizontal joint between the base panel and the foundation. The shear deformations of the structure were small despite the relatively low base moment to shear ratio. The shear slip at the horizontal joints was also small, and the crushing of the concrete at the wall toes did not result in a disproportional increase in slip under increased drift demand.

The measurements from the 3D-DIC system, which provided full-field information for the behavior of the wall at the base, compared favorably with visual observations and with data from conventional sensors. The 3D-DIC strain maps accurately depicted the cracking of the wall and provided information on the neutral axis depth and the extent of significant compression deformations at the wall toes (i.e., plastic hinge height). Furthermore, the analytical models were able to replicate the overall lateral load-displacement history and gap opening/closing behavior of the wall as well as the behavior of the PT steel. The simple finite-element model utilized in the research can serve as an effective design tool for engineers to use in practice. However, to capture the reversed-cyclic behavior of the structure, a more detailed fiber-element model was necessary.

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